Microlocal Morse Theory and Truncated Microsupport

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Preface

In these notes, We study microlocal Morse theory, which is developed by Guillermou, Kashiwara, and Schapira in [ST92], but replace microsupports with truncatd microsupprts introduced by Kashiwara, Monteiro-Fernandes, and Schapira in [KMS03].

1 Notations

We consider a commutative unital ring $\mathbf{k}$ of finite global dimension (e.g. $\mathbf{Z}$ or a field). Let X be a real analytic manifold. We denote by $\mathsf{D}^{\mathrm{b}}(\mathbf{k}_X)$ the bounded derived category of sheaves of $\mathbf{k}$-modules on X.

2 Truncated Microsupport

Definition 2.1. Let $F \in \mathsf{D}^{\mathrm{b}}(\mathbf{k}_X)$. The closed conic subset $\mathrm{SS}_k(F)$ of T^*X is defined by: $p \notin \mathrm{SS}_k(F)$ if and only if there exists an open conic neighborhood U of p such that for any $x \in \pi(U)$ and for any real-valued C^1-function φ defined on a neighborhood of x such that $\varphi(x) = 0$, $d\varphi(x) \in U$, one has

$$H^j_{\{\varphi \geqq 0\}}(F)_x = 0 \quad \text{for any } j \leqq k. \tag{2.1}$$

2.1 Functorial Properties

Proposition 2.2. Let $f\colon X \to Y$ be a morphism of real analytic manifolds and let $F \in \mathsf{D}^{\mathrm{b}}(\mathbf{k}_X)$ such that f is proper on $\mathrm{supp}(F)$. Then for any $k \in \mathbf{Z}$,

$$\mathrm{SS}_k(\mathrm{R}f_*F) \subset f_\pi f_d^{-1}(\mathrm{SS}_k(F)). \tag{2.2}$$

The equality holds in case f is a closed embedding.

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3 Lemmas from [KS90]

Consider a projective system of abelian groups $(M_n, \rho_{n,p})_n$ indexed by $\mathbf{N}$. One says that this system satisfies the Mittag-Leffler condition (ML for short) if for any $n \in \mathbf{N}$, the decreasing sequence $(\rho_{n,p}(M_p))_p$ of subgroups of M_n is stationary.

For a projective system of abelian groups $(M_n)_n$, we set

$$M_\infty := \varprojlim_n M_n.$$

Consider a projective system of complexes

$$E_n^\bullet \colon \ \cdots \to M_n^{j-1} \to M_n^j \to M_n^{j+1} \to \cdots\,, \tag{3.1}$$

and projective limit

$$E_\infty^\bullet \colon \ \cdots \to M_\infty^{j-1} \to M_\infty^j \to M_\infty^{j+1} \to \cdots\,. \tag{3.2}$$

Denote by

$$\Phi_k \colon H^k(E_\infty^\bullet) \to \varprojlim_n H^k(E_n^\bullet) \tag{3.3}$$

the natural morphism.

Proposition 3.1 ([KS90, Proposition 1.12.4])**.** Assume that for each $j \in \mathbf{Z}$, the system $\left(M_n^j\right)_n$ satisfies the ML condition. Then

(a) for each $k \in \mathbf{Z}$, the map Φ_k is surjective,

(b) if moreover, for a given i the system $\left(H^{i-1}(E_n^\bullet)\right)_n$ satisfies the ML condition, then Φ_i is bijective.

The following is called Kashiwara's lemma.

Proposition 3.2 ([KS90, Proposition 1.12.6])**.** Let $(X_s, \rho_{s,t})$ be a projective system of sets indexed by $\mathbf{R}$. Assume that for each $s \in \mathbf{R}$, the canonical maps

$$\lambda_s \colon X_s \to \varprojlim_{r<s} X_r$$

and

$$\mu_s \colon \varinjlim_{t>s} X_t \to X_s$$

are both injective (resp. surjective). Then all the maps ρ_{s_0,s_1} $(s_0 \leqq s_1)$ are injective (resp. surjective).

4 Results

Theorem 4.1. Let F in $\mathsf{D}^{\mathrm{b}}(\mathbf{k}_M)$, and $\psi \colon M \to \mathbf{R}$ be a real-valued function of class C^1, proper on $\operatorname{supp}(F)$. Let $a < b$ be real numbers. We assume $d\psi(x) \notin \mathrm{SS}_k(F)$ for $a \leqq \psi(x) < b$. For $t \in \mathbf{R}$, we set $M_t := \psi^{-1}\left(]-\infty, t[\right)$. Then, the restriction morphisms

$$H^j(M_b; F) \to H^j(M_a; F)$$

are isomorphic for each $j \leqq k$.

Lemma 4.2. Let $-\infty < a < b < +\infty$ be real numbers and $G \in \mathsf{D}^{\mathrm{b}}(\mathbf{R})$. Let k be an integer. If

$$\left(H^j_{[t,+\infty[}(G)\right)_t \cong 0 \quad \text{for all } t \in [a,b[\ , \text{ and all } j \leqq k.$$

Then one has

$$H^j(]-\infty,b[\,;G) \xrightarrow{\sim} H^j(]-\infty,a[\,;G)$$

for all $j < k$.[1]

Proof. Let $t \in [a,b[$. We will denote by I_t the open interval $]-\infty,t[$ and denote by Z_t the closed interval $]-\infty,t]$ $(=\overline{I_t})$. By applying to the distinguished triangle

$$\mathrm{R}\Gamma_{\mathbf{R}-I_t}(G) \to G \to \mathrm{R}\Gamma_{I_t}(G) \xrightarrow{+1}$$

the functor $(\ \cdot\ )_{\overline{I_t}}$, we have

$$(\mathrm{R}\Gamma_{[t,+\infty[}(G))_{\{t\}} \to G_{Z_t} \to \mathrm{R}\Gamma_{I_t}(G) \xrightarrow{+1}.$$

Aplying the functor $\mathrm{R}\Gamma(\mathbf{R};\ \cdot\ )$ to this, we have

$$(4.1) \qquad (\mathrm{R}\Gamma_{[t,+\infty[}(G))_t \to \mathrm{R}\Gamma(Z_t;G) \to \mathrm{R}\Gamma(I_t;G) \xrightarrow{+1}.$$

Then consider the following conditions:

$(a)^j_s$ $\qquad \varinjlim_{t>s} H^j(I_t;G) \xrightarrow{\sim} H^j(I_s;G) \quad$ for all $s \in [a,b[$,

$(b)^j_t$ $\qquad \varprojlim_{s<t} H^j(I_t;G) \xleftarrow{\sim} H^j(I_s;G) \quad$ for all $t \in]a,b]$.

By hypothesis, it follows from (4.1) that $H^j(Z_t;G) \xrightarrow{\sim} H^j(I_t;G)$ for $t \in [a,b[$. Similarly, we have injections $H^k(Z_t;G) \hookrightarrow H^k(I_t;G)$ for $t \in [a,b[$. Since $\varinjlim_{t>s} H^j(I_t;G) \cong H^j(Z_s;G)$, $(a)^j_s$ holds for any $j < k$. [2]Moreover, $(b)^j_t$ holds for $j \ll 0$. Let us argue by induction on j. Assume $(b)^j_t$ holds for all $t \in]a,b[$ and all $j \leqq j_0$. Applying Proposition 3.2, we find the isomorphisms

$$(4.2) \qquad H^j(I_s;G) \xrightarrow{\sim} H^j(I_t;G) \quad \text{for all } j \leqq j_0 \text{ and all } a < s \leqq t < b.$$

Let $G \xrightarrow{\sim} G^\bullet$ be a flabby resolution. Let $t \in]a,b]$. Consider a complex

$$E^\bullet_n = \Gamma(I_{t-\frac{1}{n}};G^\bullet).$$

Since $G^\bullet$ is flabby, the projective system $(E^j_n)_n$ satisfies the ML condition for all $j < k$.

[1]For k, this morphism might be injective. (not written in this draft.)

[2]Similarly,

$$\varinjlim_{t>s} H^k(I_t;G) \hookrightarrow H^k(Z_s;G)$$

gives injections

$$\varinjlim_{t>s} H^k(I_t;G) \hookrightarrow H^k(I_s;G)$$

for all $s \in [a,b[$.

By (4.2), the projective system $(H^{j_0}(E_n^\bullet))_n$ satisfies the ML condition (because we have $a < t - \frac{1}{n} < t$ for $n \gg 0$). Applying Proposition 3.1 to $(E_n^\bullet)_n$, it follows that $(b)_t^j$ holds for $j_0 + 1$ and the induction proceeds. More precisely, we have

$$\begin{aligned} H^{j_0+1}(I_t; G^\bullet) &\xrightarrow[\Phi_{j_0+1}]{\sim} \varprojlim_n H^{j_0+1}(I_{t-\frac{1}{n}}; G^\bullet) \\ &\cong \varprojlim_n H^{j_0+1}(I_{t-\frac{1}{n}}; G) \\ &\cong \varprojlim_{s<t} H^{j_0+1}(I_s; G). \end{aligned}$$

Again by Proposition 3.2, we get the isomorphisms

$$H^j(I_s; G) \xrightarrow{\sim} H^j(I_t; G) \quad \text{for all } j < k \text{ and all } a < s \leqq t \leqq b.$$

Finally, using (4.1) and the hypothesis, we have the isomorphisms

$$H^j(I_a; G) \xleftarrow{\sim} H^j(Z_a; G) \xrightarrow{\sim} H^j(I_s; G) \quad \text{for all } j < k \text{ and all } a < s \leqq b.$$

□

Proof of Theorem 4.1. First, we have

$$H^j(M_t; F) \cong H^j \, \mathrm{R}\Gamma(]-\infty, t[\,; \mathrm{R}\psi_* F)$$

for $t \in I$. Since we assume that $d\psi(x) \notin \mathrm{SS}_k(F)$ for $a \leqq \psi(x) < b$, and by (2.2), the formula $\mathrm{SS}_k(\mathrm{R}\psi_* F) \subset \psi_\pi \psi_d^{-1} \mathrm{SS}_k(F)$ holds. Hence we have

$$\left(H^j \, \mathrm{R}\Gamma_{[\psi(x), +\infty[} \mathrm{R}\psi_* F\right)_{\psi(x)} = 0$$

for each $j \leqq k$. Then we can apply Lemma 4.2 and have

$$H^j(]-\infty, b[\,; \mathrm{R}\psi_* F) \xrightarrow{\sim} H^j(]-\infty, a[\,; \mathrm{R}\psi_* F)$$

for all $j < k$. Then we can conclude that

$$H^j(M_b\,; F) \xrightarrow{\sim} H^j(M_a\,; F)$$

for all $j < k$. □

References

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